



UK Health
Security
Agency

Guidance

Mycology reference laboratory: service user handbook

Updated 16 March 2023

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This publication is available at <https://www.gov.uk/government/publications/mycology-reference-laboratory-mrl-service-user-handbook/mycology-reference-laboratory-service-user-handbook>

The Mycology Reference Laboratory (MRL) is situated at the UK Health Security Agency (UKHSA) South West Laboratory (<https://www.gov.uk/guidance/south-west-public-health-laboratory-services>) in Bristol. The laboratory provides a comprehensive service for the diagnosis and management of fungal infections through the provision of specialist laboratory services and expert clinical and technical advice.

The MRL also houses the National Collection of Pathogenic Fungi (<https://www.culturecollections.org.uk/collections/ncpf.aspx>) (NCPF), which is the only culture collection in the UK specialising in fungi pathogenic to humans and animals.

Available services

Clinical services

We provide advice by telephone to ensure selection of the most appropriate test or specimen types for particular patients, and assist with the interpretation of results.

We can also provide guidance on the clinical management of individual cases.

Reference services

Identification services, which include:

- moulds (including dermatophytes)
- unusual yeasts

- dimorphic fungal pathogens

Serological tests, examining the serology of non-indigenous mycoses.

These include:

- histoplasmosis
- coccidioidomycosis
- blastomycosis
- paracoccidioidomycosis

Reference antifungal drug susceptibility testing of yeasts and moulds – carried out as confirmatory tests for isolates producing equivocal or unexpected results.

Histological examination of referred slides.

Referred tests (charged)

Identification and antifungal susceptibility testing

The MRL can identify yeast and mould isolates, and conduct antifungal susceptibility testing.

Antifungal drugs for susceptibility testing of yeasts include:

- anidulafungin
- amphotericin
- caspofungin
- clotrimazole
- econazole
- fluconazole
- flucytosine
- isavuconazole
- itraconazole
- ketoconazole
- micafungin

- miconazole
- nystatin
- posaconazole
- voriconazole

Antifungal drugs for susceptibility testing of moulds include:

- amphotericin
- anidulafungin
- caspofungin
- clotrimazole
- griseofulvin
- isavuconazole
- itraconazole
- micafungin
- natamycin
- posaconazole
- terbinafine
- voriconazole.

Other drugs may be available on request, please contact the MRL.

Antifungal drug assays

Available tests include:

- amphotericin
- fluconazole
- flucytosine
- isavuconazole
- itraconazole
- posaconazole
- voriconazole

Please contact the MRL prior to sending samples for amphotericin or fluconazole assays.

Serological tests

Available tests include:

- tests for antibodies to *Aspergillus* and *Candida*
- tests for *Aspergillus* (galactomannan), *Candida* (mannan) and *Cryptococcus* antigens.
- Beta 1-3, D-glucan antigen detection for invasive fungal infections

Please note that the Beta 1-3, D-glucan antigen detection test has a very high negative predictive value.

Additional tests

Further tests can be conducted. These include:

- diagnostic PCR on blood, fluids, and tissue
 - panfungal
 - *Aspergillus*-specific
 - Mucoraceous mould-specific
 - PJP-specific
 - *Candida*-specific
- microscopy and culture of clinical specimens
- typing of common fungal pathogens if indicated

Please telephone the MRL for details of environmental sampling.

Laboratory turn-around times

All reports are sent by first class mail. Urgent results can be telephoned to laboratories on request provided we have appropriate contact information.

Results are available via an electronic link, please telephone for information.

The tables below indicate the minimum and mean turn-around times for tests.

Test type	Minimum turn-around time (in days)	Mean turn-around time (in days)
Aspergillus antibodies	2	2.9
Aspergillus antigen (galactomannan)	<1	1.03
Beta 1-3, D-glucan	<1	<1
Candida antibodies	2	2.8
Candida antigen (mannan)	1	1.6
Cryptococcus antigen	<1	<1
Dimorphic serology	3	5.4
Flucytosine assay	<1	<1
Itraconazole assay	<1	<1
Mould antifungal susceptibilities	5	8.7
Mould identification	1	5
PCR (Diagnostic)	2	3.2
Posaconazole assay	<1	<1
Voriconazole assay	<1	<1
Yeast antifungal susceptibilities	2	6.3
Yeast identification	<1	1

All samples should be securely packaged, conforming to current postal regulations.

How to make the best use of our service

Request forms

Please use and fully complete the [standard request forms](https://www.gov.uk/government/collections/mycology-reference-laboratory-mrl) (<https://www.gov.uk/government/collections/mycology-reference-laboratory-mrl>).

All samples should be securely packaged, conforming to current postal regulations.

Cultures for identification and susceptibility testing

When submitting samples, you should submit pure cultures on slopes rather than plates.

If you are submitting Hazard Group 3 pathogens, please give the MRL advanced warning.

When sending Hazard Group 3 pathogens, you should only send slopes of culture, please do not wait for the organisms to start to sporulate.

Serological tests

When submitting samples:

- please supply 1 to 2ml of serum, do not send whole blood
- send samples for Beta 1-3, D-glucan determination to the MRL, within 48 hours of collection

Antifungal assays

When submitting samples:

- contact the MRL before submitting amphotericin B or fluconazole assay samples as the measurement is not often indicated
- supply 1 to 2ml serum taken at the appropriate times – and labelled accordingly
- do not use serum separator tubes with a gel plug for azole drugs

Table 2. Guidelines for antifungal drug monitoring

Sample	Monitoring period
Oral flucytosine	pre-dose and 2 hours post-dose
IV flucytosine	pre-dose and 30 minutes post-dose
Oral or IV itraconazole	pre-dose after 7 days
Oral posaconazole	pre-dose after 5 days
Oral voriconazole	pre-dose after 3 to 5 days
IV voriconazole	pre-dose after 3 to 5 days
Amphotericin	1 to 2 hours post-dose
Oral or IV isavuconazole	pre-dose after 7 days
Oral fluconazole	2 hours post-dose
IV fluconazole	30 minutes post-dose

Please ensure that all high-risk samples are clearly labelled.

Interpretation of assay results

Flucytosine

Serum samples should be taken just before a dose of flucytosine and 2 hours after an oral dose or 30 minutes after an intravenous dose.

The dose and interval should be adjusted in order to produce peak serum concentrations of about 50 to 100 mg/L – and trough concentrations of 20 to

40mg/L. Levels >100mg/L are toxic.

Itraconazole

The trough level should be maintained – or be kept above – 0.5mg/l.

Less than 0.5mg/l is a low concentration.

There may be toxicity issues at higher concentrations (>4mg/l).

Liver function tests should be monitored during prolonged courses.

Posaconazole

The absorption and metabolism of posaconazole will vary from patient to patient. Recommended trough concentrations are: 0.7 mg/l for prophylaxis and >1.0 mg/l for invasive disease.

Voriconazole

The absorption and metabolism of voriconazole will vary from patient to patient.

The voriconazole trough concentration should be maintained above 1.0 mg/l.

Outcomes for bulky or disseminated infections are better when levels are >2.0 mg/l.

Levels above 6.0 mg/l are more likely to lead to liver toxicity.

Levels above 10.0 mg/l should be avoided.

Isavuconazole

A need for drug assay has not been established.

Normal levels for isavuconazole are between 2.0 to 4.0 mg/l.

No toxic levels have been established.

The drug is slow to reach steady state.

Amphotericin B

Monitoring of blood concentrations of amphotericin B, during treatment, is seldom indicated.

The optimum serum concentrations of the drug for particular fungal infections have not been determined.

Toxicity is assessed by monitoring renal function.

Although amphotericin B is nephrotoxic, high blood concentrations do not lead to greater impairment of renal function – nor does renal failure result in higher blood concentrations.

Training courses

The MRL organises regular training courses on the identification of pathogenic fungi. Full details are available on request.

Individual training can be provided for NHS and UKHSA medical, technical and scientific staff. Please telephone or write to discuss your requirements.

External quality assurance

The MRL collaborates with the Quality Assurance Laboratory in organising the UK National External Quality Assurance Schemes (<https://ukneqas.org.uk/>) (NEQAS) in mycology.

It participates in 3 UK NEQAS schemes; 2 for antibiotic assays (flucytosine and azole drugs) and the other for immunology and immunochemistry (Aspergillus, Candida, avian allergens and farmer's lung antibody tests).

The MRL also takes part in external quality assurance schemes for the validation of *Aspergillus* antigen detection and fungal PCR (Quality Control for Molecular Diagnostics), as well as the Instand e.V. (Germany) Mycology External Quality Control scheme for mould and yeast identification, cryptococcal antigen and candida antibody detection.

The National Collection of Pathogenic Fungi (NCPF)

The MRL is part of the UKHSA Biological Resource Centre Network (<https://www.culturecollections.org.uk/news/culture-collection-news/uk-biological-resource-centre-network.aspx>). It houses the only UK microbial culture collection specialising in fungi pathogenic to humans and animals.

At present, the collection holds about 4,000 strains, many of which are available for distribution.

These include:

- 700 strains of dermatophytes and related organisms from the UK and abroad from human and animal infections including many original type strains
- 1,500 mould strains from sub-cutaneous and deep-seated human and animal infections
- 1,500 pathogenic yeast strains
- more than 250 strains of dimorphic fungal pathogens

Fungal isolates

NCPF strains are supplied for research, quality control of microbiological media, disinfectants and antifungal susceptibility tests. Current prices and conditions of sale are available on request. More information is available from the UKHSA Culture Collections (<https://www.culturecollections.org.uk/>).

DNA services

As part of an ongoing modernisation of the collection, genomic DNA is being extracted for all of the fungal cultures in the collection.

These can be made available on request.

Contact details

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